

Context Matters:

Creation of an Implementation Science Toolkit to increase engagement with EPAs in General Surgery

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Disclosures

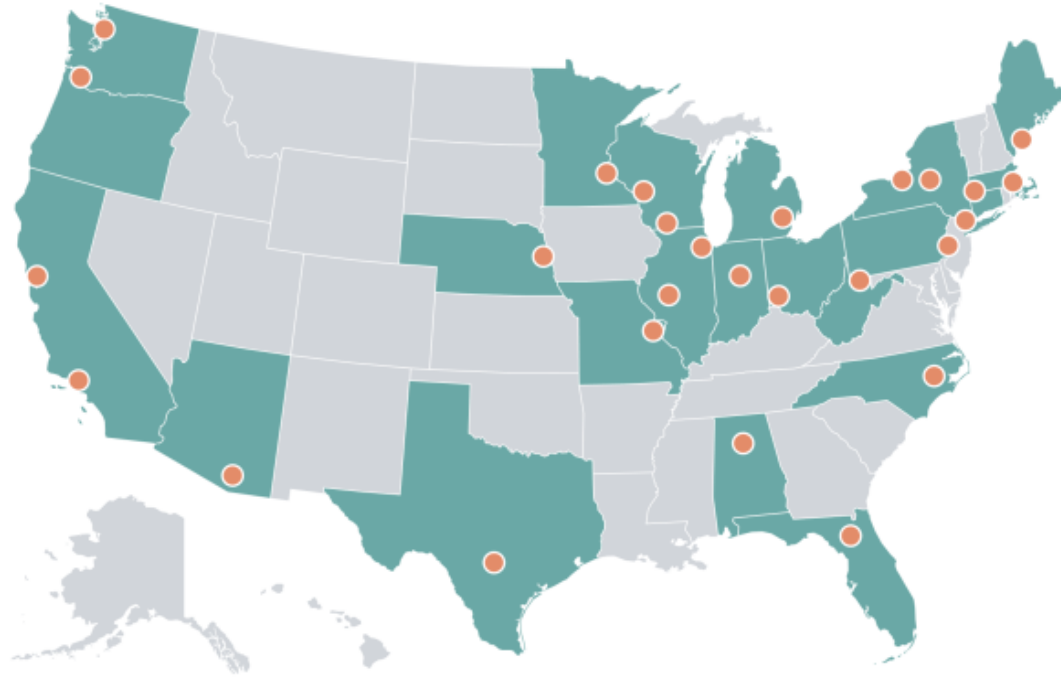
- Mark Friedell Surgical Education Research Grant, Association of Program Directors in Surgery
- Augusta Webster Education Research Grant, Feinberg School of Medicine, Northwestern University
- Faculty Scholar Research Grant, American Board of Medical Specialties and American Board of Surgery



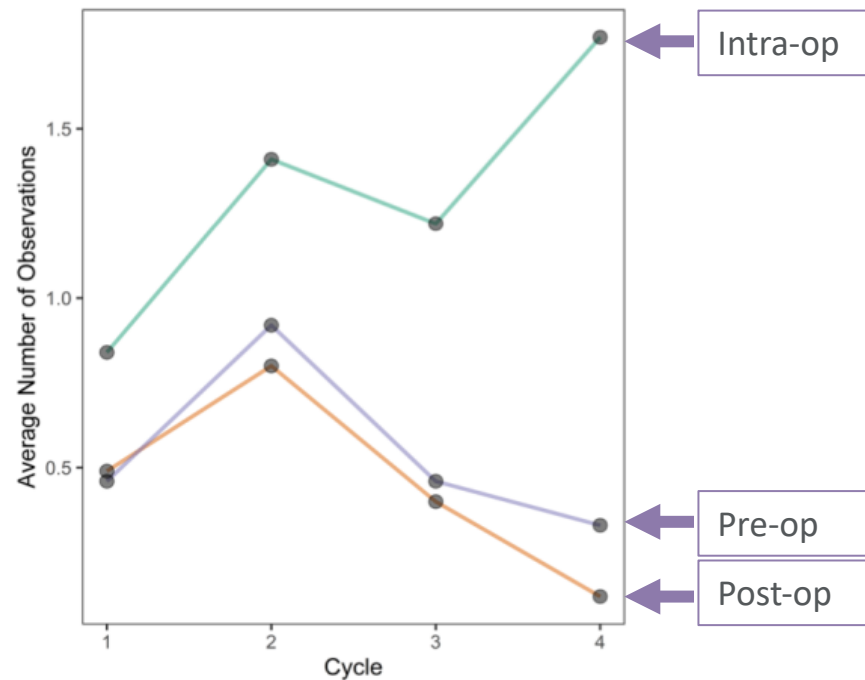
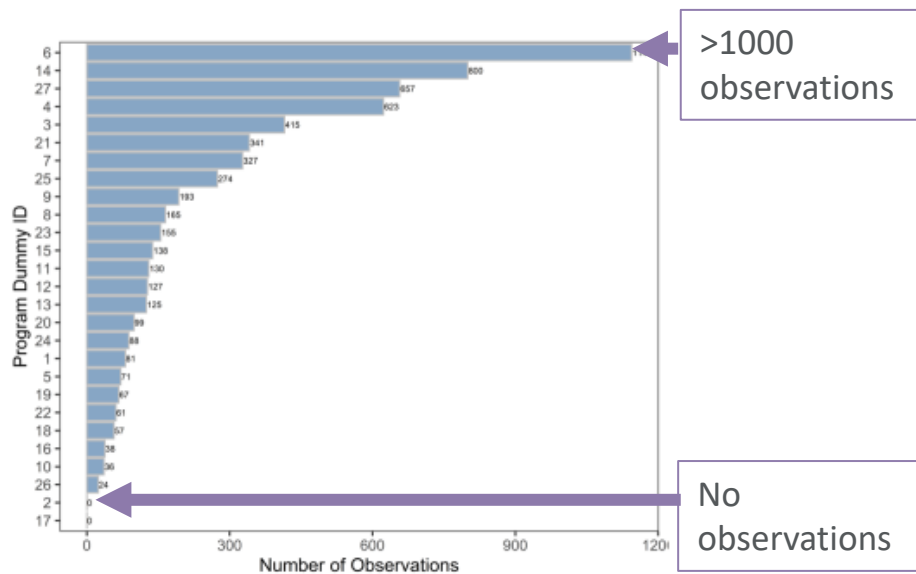
Background

- The American Board of Surgery (ABS) is moving towards competency-based assessment of surgical trainees
- The implementation of 18 core General Surgery Entrustable Professional Activities (EPAs) began July 1, 2023
- Starting in 2028, all trainees seeking general surgery board certification will be required to submit a portfolio of EPAs

The ABS conducted an EPA pilot program from 2018-2020 at 28 diverse residency programs



Engagement was highly variable



ABS EPA Best Practices Committee

- Surveys and Qualitative interviews with PDs, Faculty and Residents at participating institutions



Jacob Greenberg, MD



Faculty and resident perspectives on EPA implementation

- Faculty and Residents felt that EPAs were a **useful and valuable** addition to training
- Faculty and residents thought EPAs were **NOT overly time consuming** to complete
- Faculty and residents reported **difficulty incorporating** them into their daily workflow

(Williams-Karnesky et al, *J Surg Ed*, 2024)

ABS EPA Best Practices Committee Findings

People may **buy-in conceptually** to the idea of EPAs...
and yet may not **enact the behavior** of completing EPA evaluations

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*How do we get people to **DO THE THING**?*

Center for Dissemination & Implementation Science



**Bridging the gap between public health
knowledge and practice.**





What is implementation science?

Implementation Science is the **scientific study** of **methods and strategies** that **facilitate the uptake** of evidence-based practice and research into **regular use**.



Implementation science in (very) non-technical language

The intervention/practice/innovation is **THE THING**

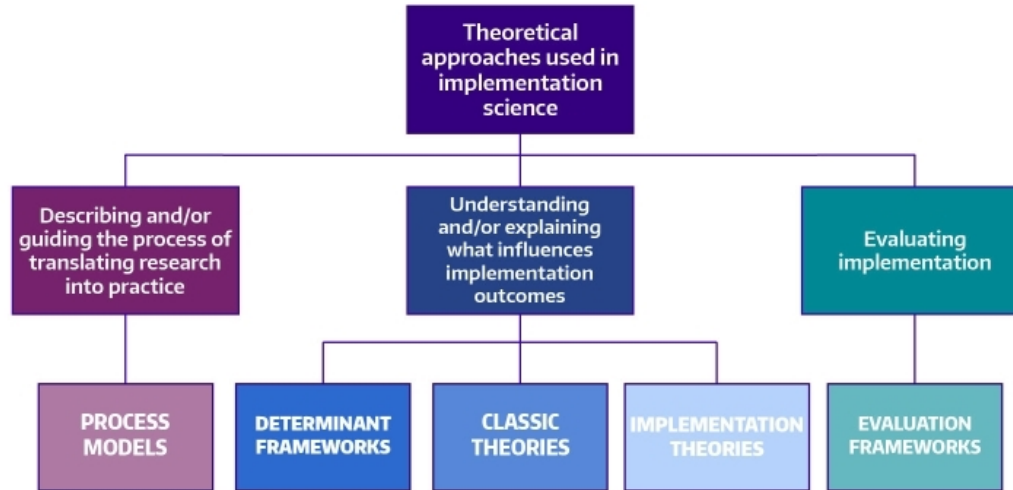
Implementation research looks at how best to help people **DO THE THING**

Implementation strategies are the stuff we do to try to help people **DO THE THING**

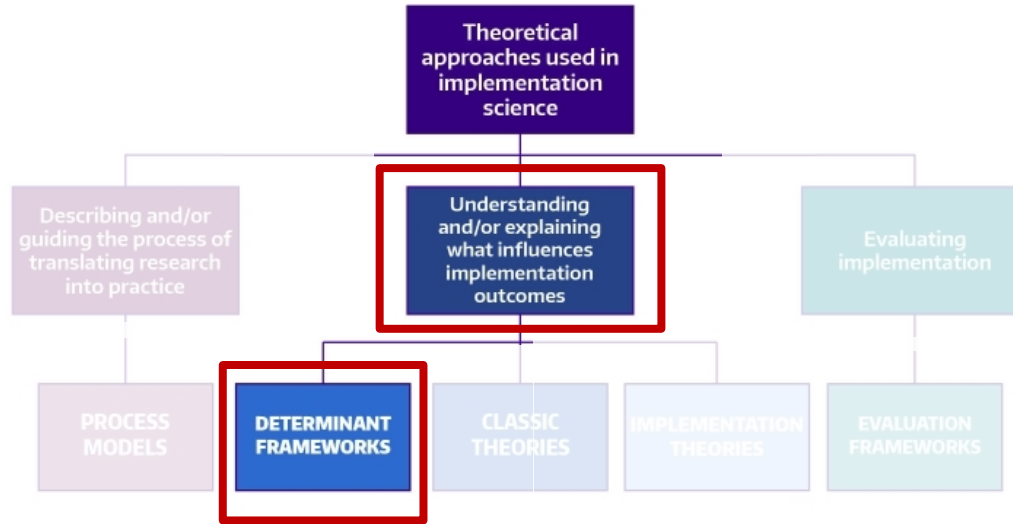
Main *implementation outcomes* are **HOW MUCH** and **HOW WELL** people **DO THE THING**

(Curran, *Implementation Science Comm*, 2020)

There are many Implementation Science strategies



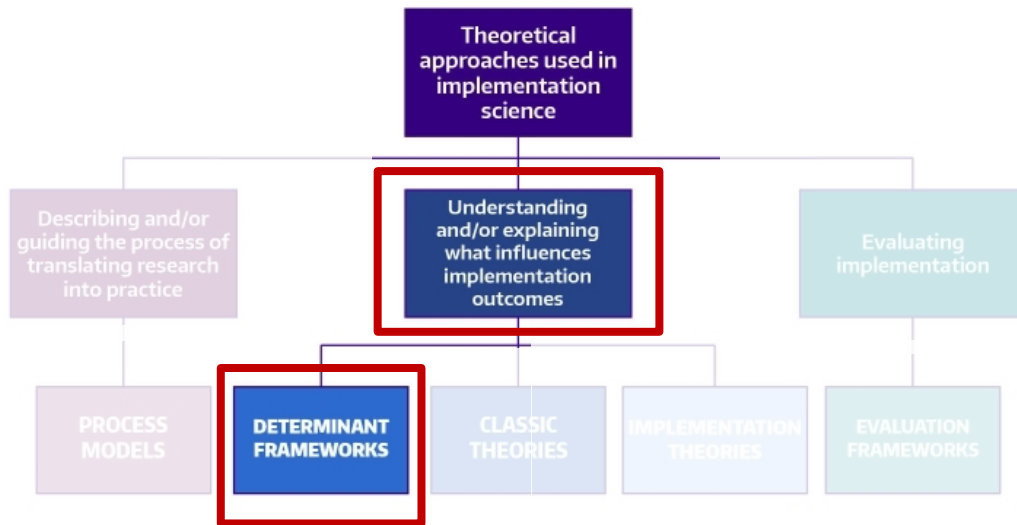
Determinant frameworks identify factors (barriers and facilitators) that influence implementation outcomes



(Nilsen, *Implementation Science*, 2015)

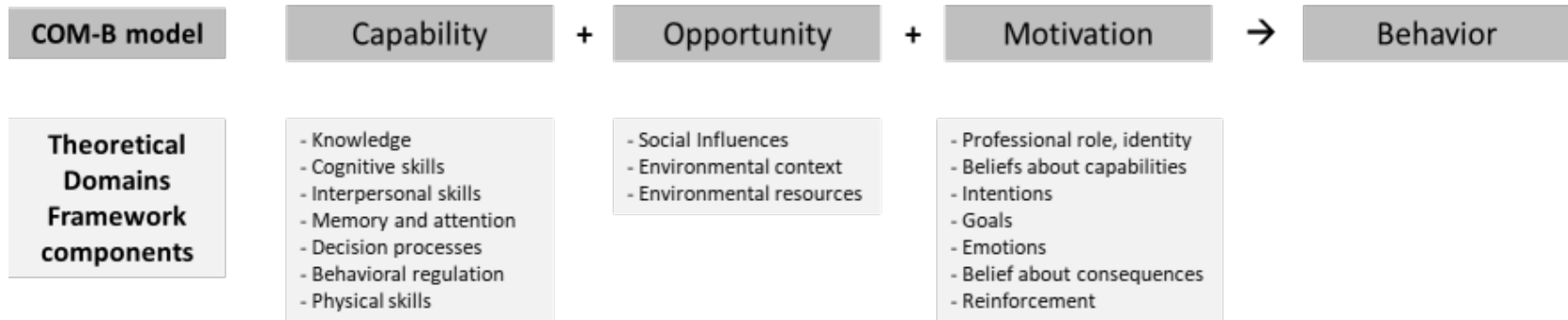
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HOW MUCH and HOW WELL people DO THE THING



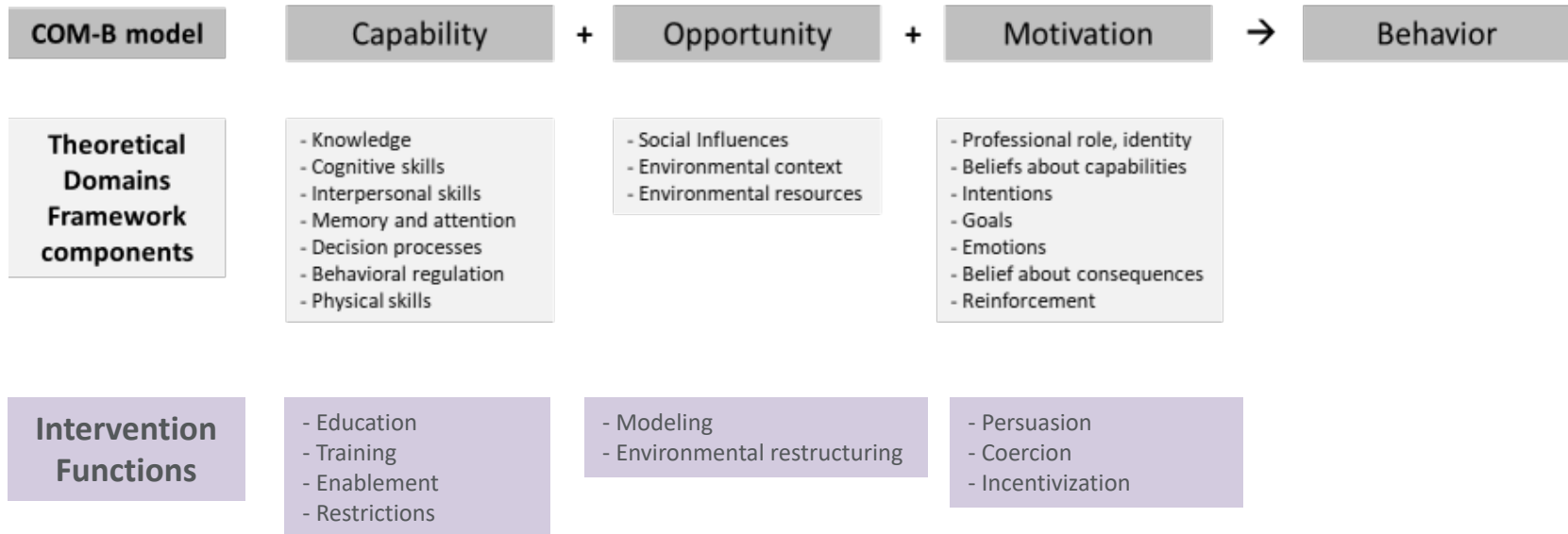
(Nilsen, *Implementation Science*, 2015)

Theoretic Domains Framework



(Atkins et al, *Implementation Science*, 2017)

Theoretic Domains Framework



(Mitchie et al, *Implementation Science*, 2011)

Using IS to increase uptake of EPAs



Survey instrument



Deductive
interview guide



Use an IRLM to link
barriers to intervention



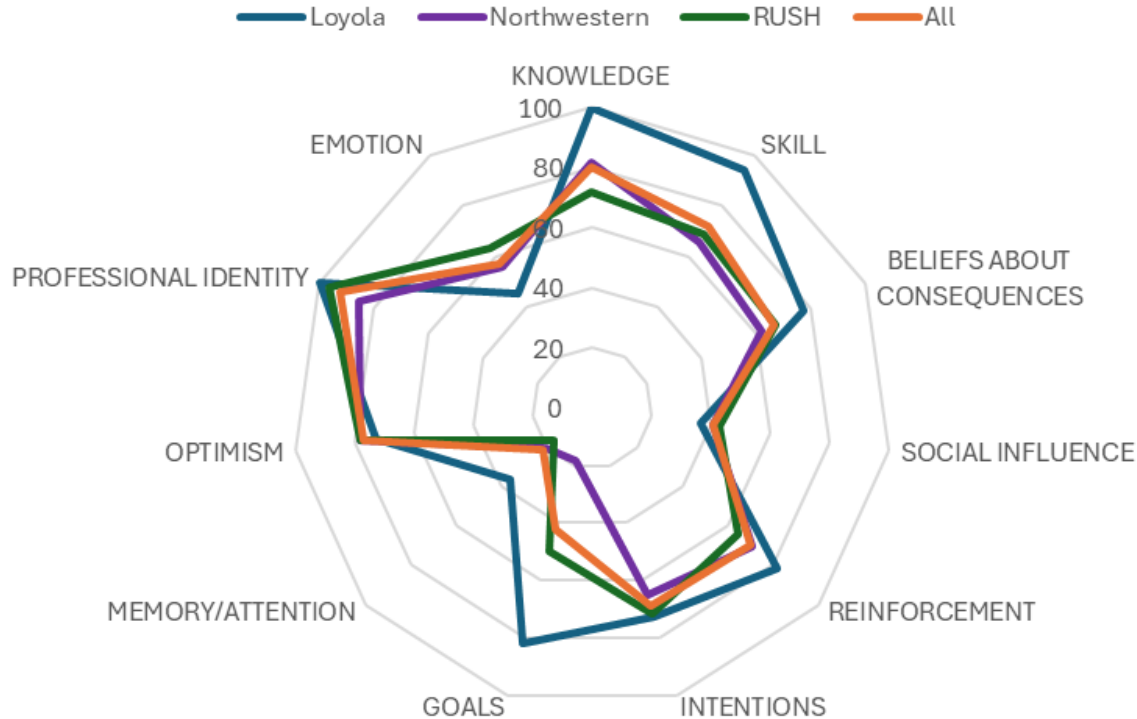
(Cohen et al Williams-Karnesky, *J Surgical Education*, 2026)



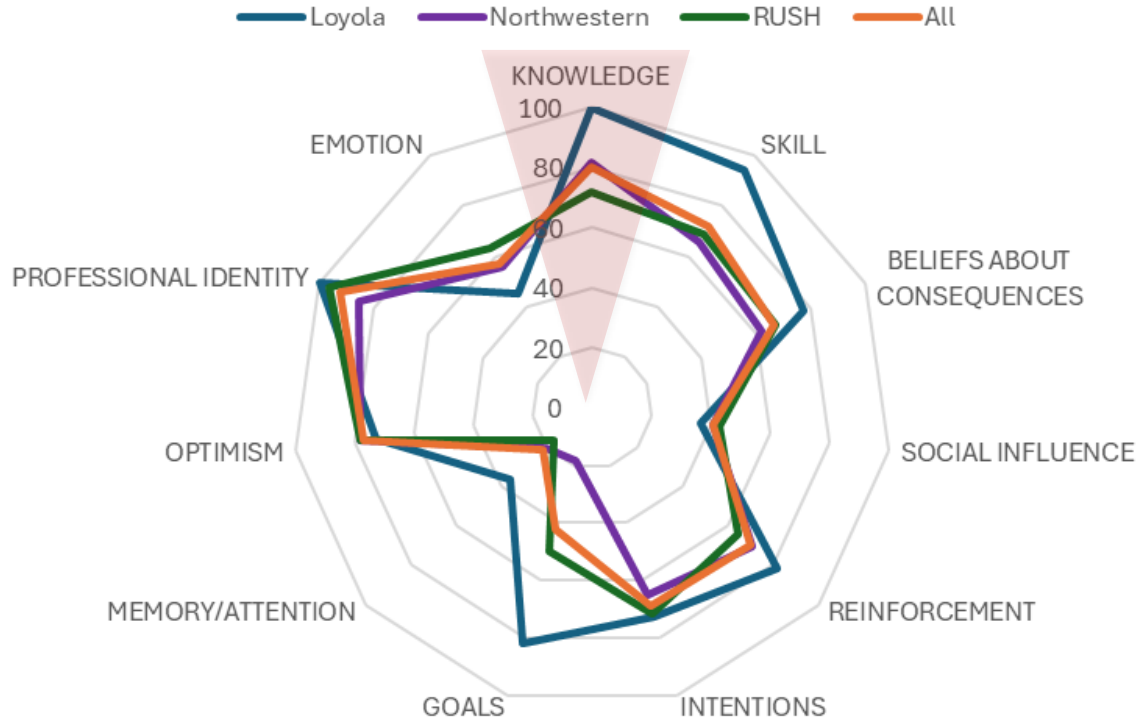
Multi-institutional study of EPA utilization

- Five Chicago-area general surgery residency programs
- Survey of faculty and residents using the TDF to identify barriers and facilitators to EPA completion
- Focus groups to link barriers to interventions
- Continuous evaluation of EPA completion

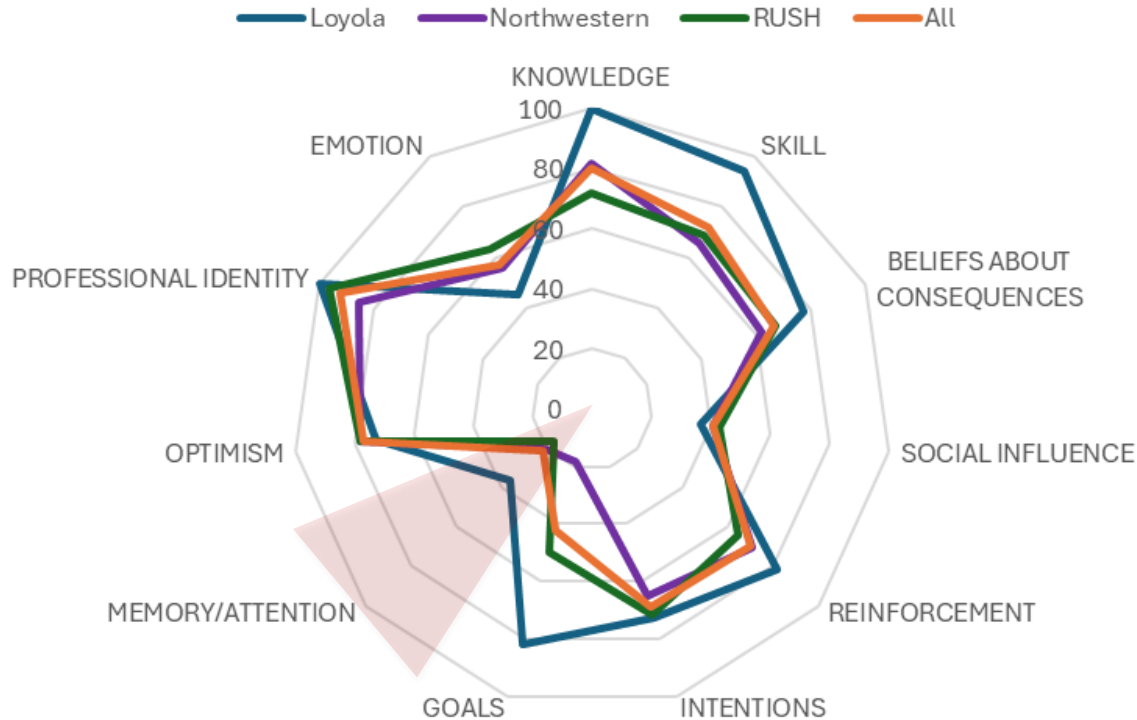
There is variation in barriers and facilitators between institutions



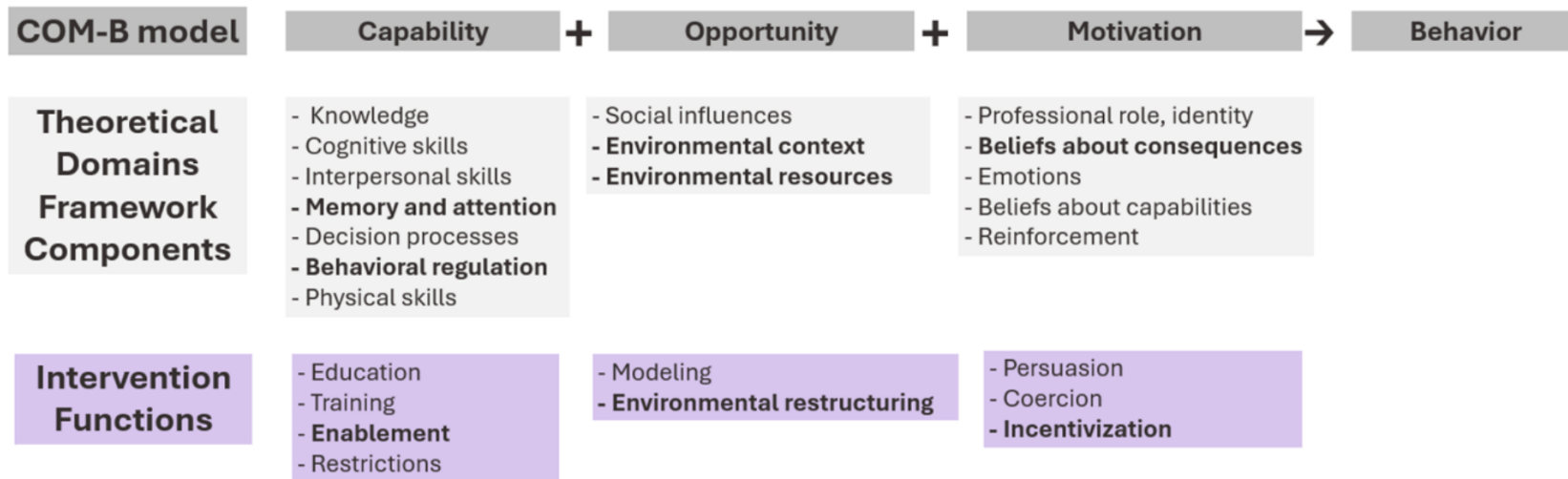
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Linking barriers to interventions



Linking barriers to interventions

- Case study: Northwestern University
 - **Knowledge:** Increased education about EPAs - grand rounds, division meetings, individual training
 - **Memory/Attention:** Provided cues for faculty to complete EPAs
 - **Reinforcement:** Incentivized faculty EPA completion by linking to compensation

Dear Program Leadership:

The EPA numbers for Q3 of the 2025-2026 academic year are in and **your program is in the top 25% of programs with an increased number of EPA assessments completed per resident compared to last quarter!**



Figure 1: Comparison of the three institutions' responses to the 10 items of the questionnaire. The chart shows scores for Institution 1 (blue), Institution 2 (orange), and Institution 3 (green) across 10 dimensions. Institution 1 consistently scores highest, particularly in SKILL and MEMORY/ATTENTION, while Institution 2 shows a notable dip in BEHAVIORAL REGULATION.

Item	Institution 1	Institution 2	Institution 3
SKILL	95	85	80
MEMORY/ATTENTION	85	75	70
BEHAVIORAL REGULATION	35	10	20
CONTEXT	30	20	25
RESOURCES	25	15	20
SOCIAL INFLUENCE	20	10	15
BELIEFS ABOUT CONSEQUENCES	15	10	15
OPTIMISM	20	15	20
REINFORCEMENT	40	35	40
PROFESSIONAL IDENTITY	60	55	55



Karen Brasel
MD, MPH

Select your institutional barriers to discover potential interventions



Refined survey and interview guide

- Survey decreased in length from 27 to 15 questions (*5 demographic + 10 Domain-specific*)

4. (*Domain: Knowledge*) **How familiar are you with the General Surgery EPAs?** (Select one: Very familiar/Somewhat familiar/Not very familiar/Not at all familiar)
5. (*Domain: Skills*) **How comfortable do you feel completing an EPA assessment on the available platform at your institution?** (Select one: Very comfortable/Somewhat comfortable /Not very comfortable /Not at all comfortable)

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- Interview Guide refined to focus on 7 key domains using an Appreciative Inquiry Paradigm (*Optimism, Beliefs about Consequences, Intention, Knowledge, Capability, Social influence, Environmental Context*)

Domain: Social Influence

How have others (e.g. peers) helped you achieve success in using EPAs?

- Follow-up prompt: What could you do to help others achieve success? What could department leaders do to help you or others achieve success?

Linking implementation strategies to barriers



Knowledge

The awareness of something and understanding key information about it.



Skills

The ability to do something, based on practice, learning or experience.



Memory

The ability to retain and remember information when it's needed.



Attention / Decision Overload

The ability to focus selectively on or be decisive about something.



Habits

A practice or behavior that someone often does automatically/without thinking about it.



Confidence in Capabilities

Someone's belief in their ability to do something.



Role or Identity

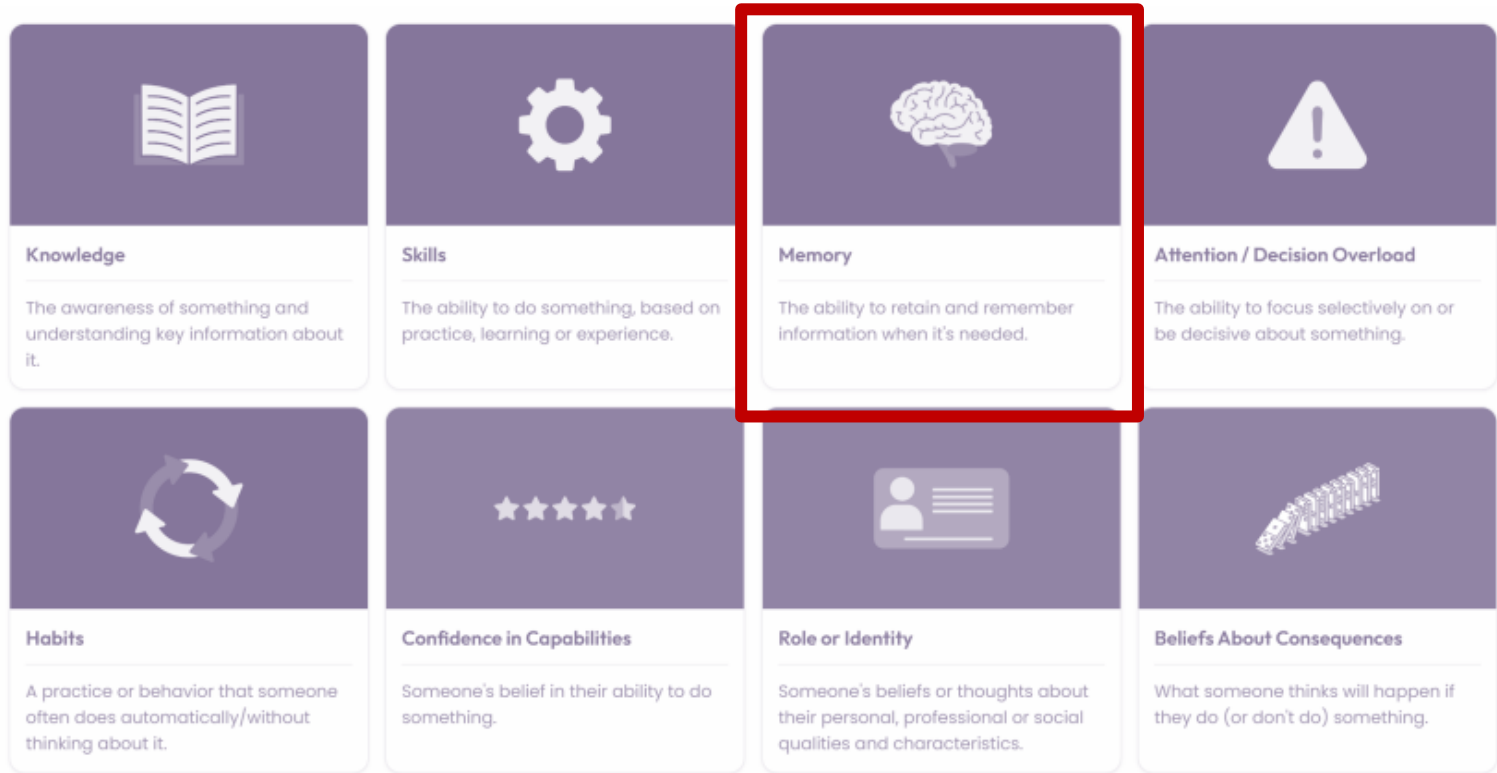
Someone's beliefs or thoughts about their personal, professional or social qualities and characteristics.



Beliefs About Consequences

What someone thinks will happen if they do (or don't do) something.

Barrier: Memory



Possible Interventions



Memory

The ability to retain and remember information when it's needed.



Improve Or Restructure Environment Or Organization

☒ Select

When existing processes, structures, workflows, or environments are challenges.

☐ Don't select

If existing processes, structures, and workflows already support the initiative, but there are knowledge or skill gaps, a lack of motivation, a lack of leadership support, or a lack of engagement.



Strengthen Data And Information Systems

☒ Select

When teams lack timely, meaningful information or data about progress, outcomes, or implementation quality.

☐ Don't select

If relevant data already exist and are being used, and the challenge is decision-making, interpretation, authority, or follow-through.

Possible Interventions – criteria to consider



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Intervention Details



Memory

Improve or restructure environment or organization

To modify processes, workflows, or physical/environmental structures.

Incorporate feedback into regular workflow

Establishing systems and processes to routinely collect, analyze, and act on feedback as part of everyday activities. This ensures continuous improvement, enhances responsiveness, and normalizes a culture of learning and adaptation within the setting.



Provide reminders

Develop reminder systems to help providers recall information and/or prompt the performance of the WHAT.



Use prompts or cues

Develop systems to help individuals recall information and/or prompt them to use the WHAT.



Toolkit validation

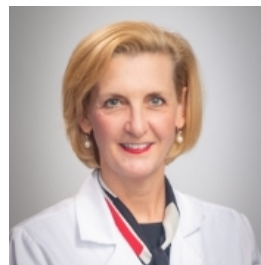
- Currently using the modified survey and interview guide to evaluate the implementation of the Complex General Surgical Oncology EPAs



Heather Lillimoe
MD



Abbey Fingeret
MD, MHPTT, PhD



Bridget Fahy
MD



Key takeaways

- Context Matters!
- IS can provide tools for identifying barriers and facilitators to behavior, and developing intervention strategies
- Even with a toolkit, this is an iterative process

**Many thanks to my many mentors,
collaborators and mentees!!**



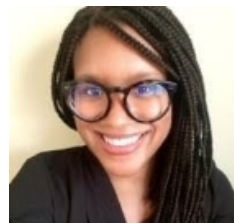
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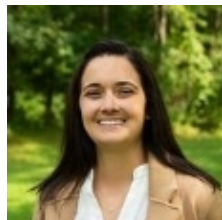
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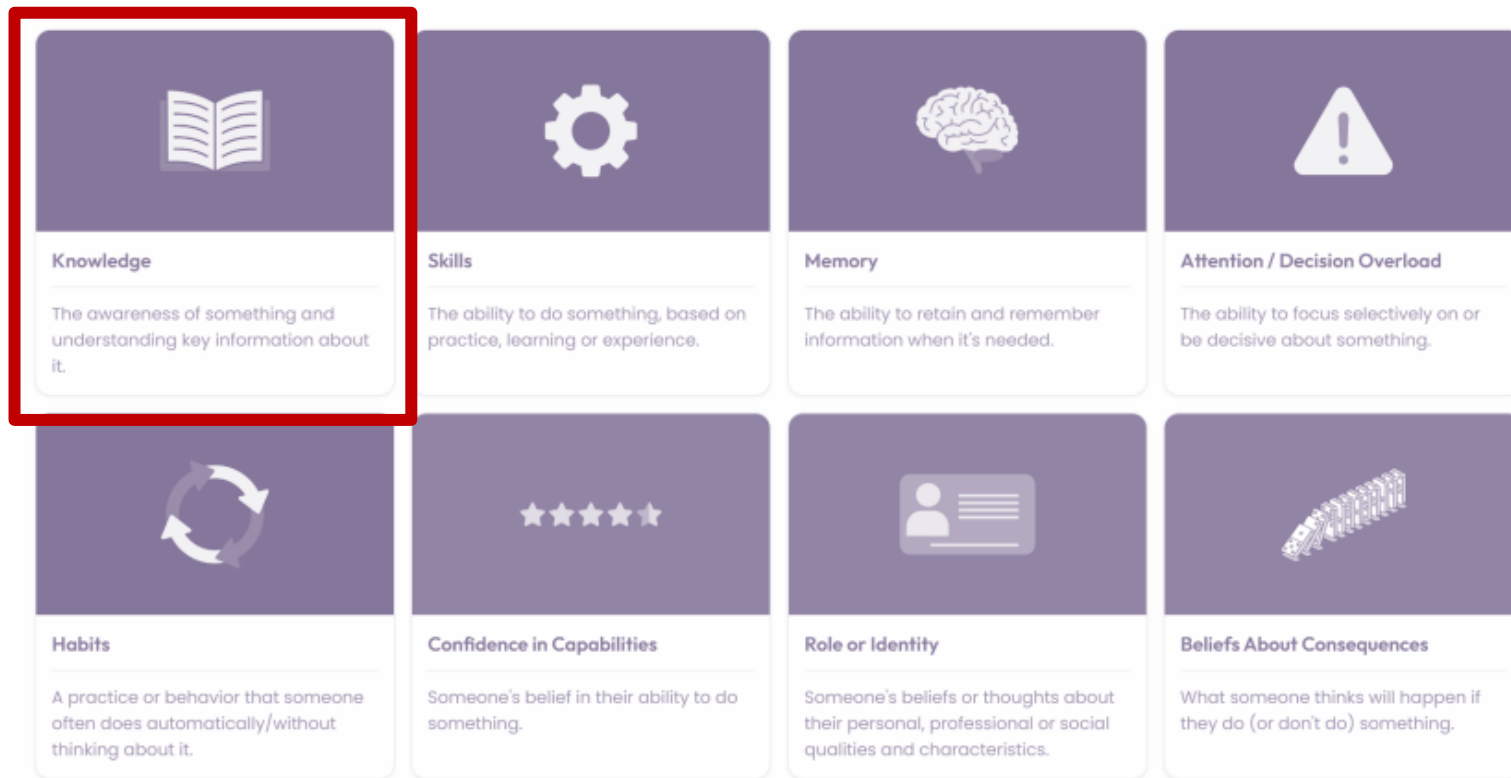
Thank You!

Please reach out with questions or comments:

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Additional Slides

AIM 2: Link implementation strategies to barriers



AIM 2: Link implementation strategies to barriers



Knowledge

The awareness of something and understanding key information about it.



Build Knowledge And Skills



Select

When people lack awareness, knowledge, or skills.



Don't select

If people already know what to do and how to do it, but lack motivation, lack leadership support, are disconnected, have competing priorities, or have structure challenges.



Communicate And Disseminate



Select

When people lack awareness, are confused, or are receiving inconsistent messages due to access or communication mechanisms.



Don't select

If the issue is motivational, people don't agree, people are not engaged, people are disconnected, there is a lack of trust, or there are issues with the decision-making processes.

AIM 2: Link implementation strategies to barriers



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AIM 2: Link implementation strategies to barriers



Knowledge

Build knowledge and skills

To increase understanding, competence, and confidence.

Audit and feedback

Measuring someone's behavior (audit) and comparing and sharing it to standards or peers (feedback).

Educational materials

Create and distribute educational materials (e.g., guidelines, manuals, and toolkits) in person, by mail, and/or electronically to improve knowledge about the WHAT.

Individual training

Providing learning opportunities to equip individuals with the specific knowledge, skills, and competencies required to perform the WHATs in their roles effectively.

Peer-to-peer learning

Partner individuals (e.g., in similar roles, occupations, etc.) for the purpose of shared training and building capacity.

Storytelling

Understanding and communicating experiences in ways that align with how humans think and feel, and that tap into people's beliefs and values. Storytelling can help illustrate the "why" of change, and can demonstrate the experience of change.

Current Research: EPA Implementation Science toolkit

- **AIM 1:** Refine the current IS survey and deductive interview guide to facilitate ease of use across institutions.
- **AIM 2:** Determine which implementation strategies best address key barriers and facilitators.
- **AIM 3:** Disseminate the IS Toolkit for EPA Implementation via an interactive platform on the ABS website.



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